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United States Patent	12396622
Kind Code	B2
Date of Patent	August 26, 2025
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Endoscope

Abstract

Provided is an endoscope capable of improving operability of the endoscope. An endoscope (10) includes: an observation optical system (76) that is disposed on a wall portion (68) of a distal end portion main body (32) adjacent to an elevator housing portion (66) and that observes a visual field region in a first direction in which the elevator housing portion (66) is open; an outflow port (45) that is provided in the distal end portion main body (32) and through which liquid flows out; and a cap (34) that is attachably and detachably attached to the distal end portion main body (32) and that has a liquid guiding surface that guides the liquid flowing out from the outflow port (45) in a direction toward the visual field region in a state where the cap (34) is attached to the distal end portion main body (32).

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Appl. No.:	17/392267
Filed:	August 03, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210361145 A1	Nov. 25, 2021

Foreign Application Priority Data

JP	2019-037792	Mar. 01, 2019
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Related U.S. Application Data

Publication Classification

Int. Cl.: **A61B1/00** (20060101); **A61B1/018** (20060101); **A61B1/12** (20060101)

U.S. Cl.:

CPC **A61B1/00101** (20130101); **A61B1/00091** (20130101); **A61B1/00098** (20130101);
A61B1/018 (20130101); **A61B1/126** (20130101);

Field of Classification Search

CPC: A61B (1/00101); A61B (1/00091); A61B (1/00098); A61B (1/018); A61B (1/126); A61B (1/00177); A61B (1/0676)

USPC: 600/107

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation of PCT International Application No. PCT/JP2020/008155 filed on Feb. 27, 2020 claiming priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2019-037792 filed on Mar. 1, 2019. Each of the above applications is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an endoscope, and particularly to an endoscope comprising, at a distal end portion of an insertion part, an elevator that changes a lead-out direction of a treatment tool and a jetting portion that jets liquid toward a visual field region of the endoscope.

2. Description of the Related Art

(2) In an endoscope, various treatment tools are led in from a treatment tool lead-in port provided in a hand operation part, and the treatment tools are led out to the outside from an elevator housing portion that is open at a distal end portion of an insertion part and are used for a treatment. For example, a treatment tool such as forceps or a contrast tube is used for a duodenoscope, and a treatment tool such as a puncture needle is used for an ultrasonic endoscope.

(3) In such a treatment tool, in order to provide a treatment at a desired position in a subject, it is necessary to change a lead-out direction of the treatment tool led out from the elevator housing portion. Therefore, an elevator (also referred to as a raising base) is rotatably provided in the elevator housing portion, and a treatment tool operating member that changes a posture of the elevator between an elevating position and a lying position is provided in the hand operation part (see JP2015-29764A).

(4) JP2015-29764A discloses an endoscope having a liquid jetting device that jets liquid toward a visual field region of the endoscope.

(5) The endoscope of JP2015-29764A has a raising base housing recessed portion that is recessed in a vicinity of a distal end portion of an insertion part with an opening surface facing sideways, a raising base supported in the raising base housing recessed portion so as to be raised and laid down, and a water supply pipe line that supplies liquid, and a send-out port of a distal end of the water supply pipe line is open in the raising base housing recessed portion. When the raising base is raised, a track of water sent from the send-out port of the water supply pipe line overlaps the raising base, and a direction of water supply is changed by a water supply guiding surface of the raising base.

(6) The water supply guiding surface of the raising base of JP2015-29764A has a center part formed into a deep groove shape, and this water supply guiding surface also functions as a guiding surface for leading out the treatment tool to the outside.

SUMMARY OF THE INVENTION

(7) In the field of an endoscope that uses a treatment tool, there is a demand for treating an observation site using the treatment tool while washing the observation site by jetting liquid to the observation site in order to ensure a clear image of the observation site.

(8) However, the endoscope of JP2015-29764A cannot simultaneously perform a liquid supply operation and a treatment tool lead-out operation by using the water supply guiding surface formed on the elevator. Therefore, the above-described demand cannot be satisfied, and as a result, there is a problem in that operability is poor.

(9) The present invention has been made in view of such circumstances, and an object of the present invention is to provide an endoscope capable of improving the operability of the endoscope.

- (10) In order to achieve the object of the present invention, an endoscope according to an aspect of the present invention comprises: a distal end portion main body that is provided on a distal end side of an insertion part; an elevator housing portion that is provided in the distal end portion main body and that is open in a first direction orthogonal to a longitudinal axis direction of the insertion part; an elevator that is rotatably provided in the elevator housing portion; an observation optical system that is disposed on a wall portion of the distal end portion main body adjacent to the elevator housing portion and that observes a visual field region in the first direction in which the elevator housing portion is open; an outflow port that is provided in the distal end portion main body and through which liquid flows out; and a cap that is attachably and detachably attached to the distal end portion main body and that has a liquid guiding surface that guides the liquid flowing out from the outflow port in a direction toward the visual field region.
- (11) In the aspect of the present invention, it is preferable that the cap has an inflow portion facing the outflow port in a state where the cap is attached to the distal end portion main body, a jetting portion that is open toward the visual field region, and a flow path connecting the inflow portion and the jetting portion to each other, and that a wall surface of the flow path is formed as the liquid guiding surface.
- (12) In the aspect of the present invention, it is preferable that the jetting portion is configured such that a jetting direction is directed to a position separated from the observation optical system in an optical axis direction of the observation optical system.
- (13) In the aspect of the present invention, it is preferable that a position of the jetting portion in the longitudinal axis direction is located on a base end side in the longitudinal axis direction with respect to a position of the observation optical system in the longitudinal axis direction.
- (14) In the aspect of the present invention, it is preferable that a position of the jetting portion in the longitudinal axis direction is located on a distal end side in the longitudinal axis direction with respect to a position of the observation optical system in the longitudinal axis direction.
- (15) In the aspect of the present invention, it is preferable that the elevator housing portion and the observation optical system are disposed adjacent to each other in a second direction orthogonal to the longitudinal axis direction and orthogonal to the first direction, and that the jetting portion is disposed on a side opposite to the observation optical system with the elevator housing portion interposed therebetween.
- (16) In the aspect of the present invention, it is preferable that the elevator housing portion and the observation optical system are disposed adjacent to each other in a second direction orthogonal to the longitudinal axis direction and orthogonal to the first direction, and that the jetting portion is disposed on a side opposite to the elevator housing portion with the observation optical system interposed therebetween.
- (17) In the aspect of the present invention, it is preferable that the outflow port is configured such that an outflow direction is directed in a direction along the longitudinal axis direction.
- (18) In the aspect of the present invention, it is preferable that the outflow port is configured such that an outflow direction is directed in a direction along the second direction.
- (19) In the aspect of the present invention, it is preferable that the liquid guiding surface is formed by an inner wall surface of a groove formed in the cap.
- (20) In the aspect of the present invention, it is preferable that the liquid guiding surface is formed by an inner wall surface of a through-hole formed in the cap.
- (21) According to the present invention, it is possible to improve operability of the endoscope.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a configuration diagram of an endoscope system comprising an endoscope according

to an embodiment.

(2) FIG. 2 is an enlarged perspective view of a distal end portion.

(3) FIG. 3 is an assembly perspective view showing a configuration of the distal end portion shown in FIG. 2.

(4) FIG. 4 is a perspective view of a distal end portion main body as viewed from an X(+) side.

(5) FIG. 5 is a perspective view showing assembly of a distal end portion main body and an elevator.

(6) FIG. 6 is a cross-sectional view of a main part of a distal end portion showing a configuration of a liquid jetting device of a first embodiment.

(7) FIG. 7 is a front view of the distal end portion shown in FIG. 6 as viewed from a Y(+) side.

(8) FIG. 8 is a top view of the distal end portion shown in FIG. 6 as viewed from a Z(+) side.

(9) FIG. 9 is an explanatory view showing an image of a visual field region displayed on a display.

(10) FIG. 10 is a top view of a distal end portion having a liquid jetting device of a second embodiment.

(11) FIG. 11 is a schematic cross-sectional view of the distal end portion shown in FIG. 10 as viewed from an X(+) side.

(12) FIG. 12 is a top view of a distal end portion having a liquid jetting device of a third embodiment.

(13) FIG. 13 is a top view of a distal end portion having a liquid jetting device of a fourth embodiment.

(14) FIG. 14 is a side view of a left inner wall surface of a cap shown in FIG. 13 as viewed from an X(+) side.

(15) FIG. 15 is a perspective view of a distal end portion main body comprising a liquid jetting portion.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(16) Hereinafter, preferred embodiments of an endoscope of the present invention will be described with reference to the accompanying drawings.

(17) FIG. 1 is a configuration diagram of an endoscope system 12 comprising an endoscope 10 according to an embodiment of the present invention. The endoscope system 12 comprises an endoscope 10, a processor device 14, a light source device 16, and a display 18.

(18) The endoscope 10 comprises an operation part 22 provided with an elevating operation lever 20 and an insertion part 24 provided on a distal end side of the operation part 22 and inserted into a subject.

(19) The insertion part 24 has a longitudinal axis A extending from a base end to a distal end, and comprises a flexible portion 26, a bending portion 28, and a distal end portion 30 in this order from the base end to the distal end. The detailed configuration of the distal end portion 30 will be described below, but first, a schematic configuration of the distal end portion 30 will be described.

(20) FIG. 2 is an enlarged perspective view of the distal end portion 30. Here, the endoscope 10 (see FIG. 1) of the embodiment is a side-viewing endoscope used as, for example, a duodenoscope, and the distal end portion 30 of FIG. 2 has a configuration in the side-viewing endoscope.

(21) FIG. 3 is an assembly perspective view showing the configuration of the distal end portion 30 shown in FIG. 2. As shown in FIG. 3, the distal end portion 30 has a distal end portion main body 32 and a cap 34, and is configured by mounting the cap 34 on the distal end portion main body 32. The distal end portion main body 32 is provided on a distal end side of the insertion part 24 (see FIG. 1), and the distal end portion main body 32 is provided with an elevator 36, which will be described below. In FIGS. 2 and 3, a state in which the elevator 36 is located in a lying position is shown.

(22) In addition, FIGS. 2 and 3 show various components disposed inside the insertion part 24 of the endoscope 10 (see FIG. 1). Specifically, a treatment tool channel 38 that guides a distal end portion of a treatment tool (not shown) to the distal end portion main body 32, an elevating

operation wire **40** (hereinafter, referred to as a wire **40**) for performing an operation of changing a lead-out direction of the distal end portion of the treatment tool that is led out from the distal end portion main body **32**, an air and water supply tube **42**, and a liquid supply tube **44** are provided. In addition, although not shown in FIGS. **2** and **3**, components such as an angle wire for performing an operation of changing a bending direction of the bending portion **28** (see FIG. **1**), a signal cable for transmitting an image signal, and a light guide for transmitting illumination light are also provided.

(23) In the present specification, a configuration of each part will be described by using a three-dimensional orthogonal coordinate system including three-axis directions (X-axis direction, Y-axis direction, and Z-axis direction). For example, in FIGS. **1** to **3**, a Z(+) direction indicates an upward direction and a Z(-) direction indicates a downward direction. In addition, an X(+) direction indicates a right direction and an X(-) direction indicates a left direction. Furthermore, a Y(+) direction indicates a direction toward the distal end of the distal end portion **30**, and a Y(-) direction indicates a direction toward the base end side of the distal end portion **30**. The Y-axis direction including the Y(+) direction and the Y(-) direction is parallel to the longitudinal axis A direction of the insertion part **24**.

(24) Returning to FIG. **1**, the operation part **22** has a substantially cylindrical shape as a whole. The operation part **22** has an operation part main body **46** in which the elevating operation lever **20** is rotatably provided, and a grip portion **48** connected to the operation part main body **46**, and a base end portion of the insertion part **24** is provided on the distal end side of the grip portion **48** through a bending-proof pipe **50**. The grip portion **48** is a portion that is gripped by an operator during the operation of the endoscope **10**.

(25) The operation part main body **46** comprises a universal cable **52**. A light source connector **54** is provided on the distal end side of the universal cable **52**. The light source connector **54** is provided with an electric connector **56** branched. The electric connector **56** is connected to the processor device **14**, and the light source connector **54** is connected to the light source device **16**.

(26) An air and water supply button **57** and a suction button **59** are provided on the operation part main body **46** side by side. In a case where the air and water supply button **57** is operated, air and liquid are supplied to the air and water supply tube **42** in FIG. **2**, and air and liquid can be jetted from an air and water supply nozzle **58** provided in the distal end portion main body **32**.

(27) The liquid supply tube **44** is disposed in the universal cable **52** from the insertion part **24** through the operation part **22**, and an opening at an end portion thereof is formed as a connector connection part (not shown) for water jet to the light source connector **54**. A water tank (not shown) is connected to the connector connection part through a liquid supply pump (not shown). In a case where the liquid supply pump is driven by an electric switch provided separately, water is supplied from the water tank to the liquid supply tube **44** through the connector connection part. As shown in FIG. **3**, the distal end of the liquid supply tube **44** is open as an outflow port **45** in the distal end portion main body **32**. The liquid may be supplied to the liquid supply tube **44** by switching a liquid supply pipe line from the air and water supply tube **42** to the liquid supply tube **44** by fully pressing the air and water supply button **57**.

(28) In a case where the suction button **59** in FIG. **1** is operated, body fluid, such as blood, can be sucked through the treatment tool channel **38** from a suction port that also serves as a treatment tool lead-out port **60** provided in the distal end portion main body **32** in FIG. **2**.

(29) As shown in FIG. **1**, a pair of angle knobs **62** and **62** that perform an operation of bending the bending portion **28** are disposed on the operation part main body **46**. The pair of angle knobs **62** and **62** are coaxially provided so as to be rotationally movable.

(30) The elevating operation lever **20** is rotatably provided coaxially with the angle knobs **62** and **62**. The elevating operation lever **20** is rotationally operated by a hand of the operator who grips the grip portion **48**. In a case where the elevating operation lever **20** is rotationally operated, the wire **40** in FIG. **2** is pushed and pulled in conjunction with the rotary operation of the elevating

operation lever **20**. By such an operation of the wire **40**, the posture of the elevator **36** coupled to the distal end side of the wire **40** is changed between a lying position in FIG. **3** and an elevating position (not shown).

(31) As shown in FIG. **1**, the grip portion **48** of the operation part **22** is provided with a treatment tool lead-in port **64** to which the treatment tool is led in. The treatment tool (not shown) led in from the treatment tool lead-in port **64** with the distal end portion at the head is inserted into the treatment tool channel **38** in FIG. **2** and is led out to the outside from the treatment tool lead-out port **60** provided in the distal end portion main body **32**.

(32) As shown in FIG. **1**, the flexible portion **26** of the insertion part **24** has a spiral pipe (not shown) formed by spirally winding a thin elastic strip-shaped plate made of metal. The flexible portion **26** is configured by covering the outside of the spiral pipe with a cylindrical net body braided using a metal wire and covering an outer peripheral surface of the net body with an outer sheath made of resin.

(33) The bending portion **28** of the insertion part **24** has a structure in which a plurality of angle rings (not shown) are coupled to each other so as to be rotationally moved. The bending portion **28** is configured by covering the outer periphery of the structure with a cylindrical net body braided using a metal wire and covering the outer peripheral surface of the net body with a cylindrical outer sheath made of rubber. For example, four angle wires (not shown) are disposed from the bending portion **28** configured in this way to the angle knobs **62** and **62**, and the bending portion **28** is bent upward, downward, leftward, and rightward by pushing and pulling these angle wires by the rotational operation of the angle knobs **62** and **62**.

(34) The endoscope **10** of the embodiment configured as described above is, as described above, a side-viewing endoscope used as a duodenoscope, and the insertion part **24** is inserted into a subject through the oral cavity. The insertion part **24** is inserted from the esophagus via the stomach to the duodenum, and a treatment, such as a predetermined test or a remedy, is performed.

(35) Examples of the treatment tool used in the endoscope **10** of the embodiment include biopsy forceps having a cup capable of collecting a living tissue at a distal end portion thereof, an endoscopic sphincterotomy (EST) knife, or a contrast tube.

(36) Next, the structure of the distal end portion **30** shown in FIGS. **2** and **3** will be described in detail.

(37) The distal end portion **30** comprises the distal end portion main body **32** and the cap **34** that is attachably and detachably mounted on the distal end portion main body **32**. The cap **34** is formed of a substantially cylindrical body whose distal end side is sealed and base end side is open, and an opening window **34A** having a substantially rectangular shape is provided on a part of the outer peripheral surface thereof. In a case where the cap **34** is mounted on the distal end portion main body **32** as shown in FIG. **2**, the opening window **34A** is disposed in the Z(+) direction, which is a first direction orthogonal to the longitudinal axis A direction. Thus, the treatment tool lead-out port **60** of the distal end portion main body **32** communicates with the opening window **34A** through an elevator housing portion **66**. The elevator housing portion **66** will be described below.

(38) The cap **34** is made of an elastic material, for example, a rubber material such as fluororubber or silicon rubber, or a resin material such as polysulfone or polycarbonate. A projecting engaging portion (not shown) to be engaged with a groove-shaped engaged portion (not shown) formed on the distal end portion main body **32** is provided on the base end side of the cap **34**, and the engaging portion is engaged with the engaged portion, so that the cap **34** is mounted on the distal end portion main body **32**. In a case where the treatment of the endoscope **10** is completed, the cap **34** is removed from the distal end portion main body **32** and is washed and disinfected, or is discarded as a disposable.

(39) As shown in FIG. **3**, the distal end portion main body **32** has a pair of wall portions **68** and **70** projecting toward the Y(+) side, and these wall portions **68** and **70** are disposed to face each other in the X-axis direction. In addition, the above-described elevator housing portion **66** that houses the

elevator **36** is provided between the wall portion **68** and the wall portion **70**. The elevator housing portion **66** is open in the Z(+) direction and the Z(-) direction, which are the first directions orthogonal to the longitudinal axis A direction.

(40) With regard to a positional relationship between the wall portion **68** and the elevator housing portion **66**, the wall portion **68** is disposed adjacent to the elevator housing portion **66** in the X direction. Further, an illumination optical system **74** having an illumination window and an observation optical system **76** having an observation window are disposed adjacent to each other in the Y direction on an upper surface **68A** of the wall portion **68** on the Z(+) side. With this observation optical system **76**, it is possible to observe the visual field region in the Z(+) direction in which the elevator housing portion **66** is open.

(41) With regard to a positional relationship between the observation optical system **76** and the elevator housing portion **66**, the observation optical system **76** and the elevator housing portion **66** are disposed adjacent to each other in the X direction, which is a second direction orthogonal to the longitudinal axis A direction and orthogonal to the Z(+) direction. The above-described air and water supply nozzle **58** is provided on the distal end portion main body **32** toward the observation optical system **76**, whereby the observation optical system **76** is washed and dried by air and liquid jetted from the air and water supply nozzle **58**.

(42) A housing chamber **72** is provided inside the wall portion **68**. An illumination unit (not shown) and an imaging unit (not shown) are housed in the housing chamber **72**. The illumination unit comprises a light guide (not shown) disposed on the housing chamber **72** side of the illumination optical system **74**. The light guide is disposed in the universal cable **52** from the insertion part **24** of the endoscope **10** (see FIG. **1**) through the operation part **22**, and a base end thereof is connected to the light source connector **54**. Accordingly, in a case where the light source connector **54** is connected to the light source device **16**, illumination light from the light source device **16** is emitted from the illumination optical system **74** toward the visual field region existing in the Z(+) direction through the light guide.

(43) The above-described imaging unit comprises a complementary metal oxide semiconductor (CMOS) type or charge coupled device (CCD) type imaging element (not shown) disposed on the housing chamber **72** side of the observation optical system **76**. A subject image is formed on an image forming surface of the imaging element by an image forming lens (not shown) constituting the observation optical system **76**. A distal end of a signal cable (not shown) is connected to the imaging element, the signal cable is disposed in the universal cable **52** from the insertion part **24** of the endoscope **10** (see FIG. **1**) through the operation part **22**, and a base end thereof is connected to the electric connector **56**. Accordingly, in a case where the electric connector **56** is connected to the processor device **14**, an imaging signal of the subject image obtained by the observation optical system **76** is transmitted to the processor device **14** through the signal cable. Then, the imaging signal is subjected to image processing by the processor device **14**, and then displayed on the display **18** as a subject image.

(44) Next, the configuration of the elevator **36** will be described with reference to FIGS. **4** and **5**. FIG. **4** is a perspective view of the distal end portion main body **32** of FIG. **3** as viewed from the X(+) side, and FIG. **5** is a perspective view showing assembly of the distal end portion main body **32** and the elevator **36**.

(45) An elevating lever housing chamber **78** is provided on a side surface of the wall portion **70** on the X(+) side, and an elevating lever **80** is housed in the elevating lever housing chamber **78**. The elevating lever housing chamber **78** has a fan-shaped concave shape, is covered with a protective plate (not shown), and is sealed. In addition, a through-hole **82** for allowing the wire **40** to penetrate therethrough is provided on a side surface **78A** of the elevating lever housing chamber **78** on the Y(-) side along the Y-axis direction. A distal end of the wire **40** is inserted into the through-hole **82** and is fixed to a lever portion **80A** of the elevating lever **80**.

(46) As shown in FIG. **5**, the wall portion **70** comprises a through-hole **84** penetrating through the

elevating lever housing chamber **78** and the elevator housing portion **66** along the X-axis direction. A rotary shaft **86** of the elevating lever **80** is disposed so as to penetrate through the through-hole **84** and is rotatably and pivotally supported. A distal end of the rotary shaft **86** is fitted into a hole **88** provided in a base portion **36A** of the elevator **36**. Thus, the elevating lever **80** and the elevator **36** are coupled to each other through the rotary shaft **86**.

(47) Therefore, according to the distal end portion main body **32** configured as described above, in a case where the wire **40** is pushed and pulled by the elevating operation lever **20** (see FIG. **1**), the rotary shaft **86** rotates together with the elevating lever **80**, whereby the posture of the elevator **36** is changed between the lying position and the elevating position (not shown). By this operation, the lead-out direction of the treatment tool led out from the treatment tool lead-out port **60** to the elevator housing portion **66** can be changed. An O-ring (not shown) is disposed between the rotary shaft **86** and the through-hole **84** to prevent gas and liquid from entering the elevator housing portion **66** and the elevating lever housing chamber **78**.

(48) Next, a liquid jetting device that jets liquid toward the visual field region of the observation optical system **76** will be described.

(49) FIG. **6** is a cross-sectional view of the main part of the distal end portion **30** showing the configuration of a liquid jetting device **90** of a first embodiment, and is a cross-sectional view of the main part of the distal end portion **30** as viewed from the X(-) side. FIG. **7** is a front view of the distal end portion **30** shown in FIG. **6** as viewed from the Y(+) side, and FIG. **8** is a top view of the distal end portion **30** shown in FIG. **6** as viewed from the Z(+) side.

(50) As shown in FIGS. **6** to **8**, the liquid jetting device **90** of the first embodiment has: the liquid supply tube **44** having the outflow port **45** that is open at the distal end thereof; and an inflow portion **92**, a jetting portion **94**, and a bent flow path **96** provided in the cap **34**. As shown in FIG. **6**, a distal end of the liquid supply tube **44** is connected to the outflow port **45** of the distal end portion main body **32** through a joint pipe **43**. In addition, the joint pipe **43** is adhered or screwed to the outflow port **45**.

(51) As an example, the outflow port **45** is configured in the distal end portion main body **32** such that an outflow direction is directed in a direction along the longitudinal axis A direction (see FIG. **3**).

(52) As shown in FIG. **6**, the inflow portion **92** provided in the cap **34** is provided in an end surface **34B** of the cap **34** on the Y(-) side. In addition, the inflow portion **92** is provided at a position facing the outflow port **45** in a state where the cap **34** is attached to the distal end portion main body **32**.

(53) The jetting portion **94** is provided on an upper surface **34C** of the cap **34** on the Z(+) side. In addition, a position of the jetting portion **94** in the longitudinal axis A direction is located on the base end side in the longitudinal axis A direction with respect to a position of the observation optical system **76** in the longitudinal axis A direction. Further, the jetting portion **94** is open toward a visual field region C of the observation optical system **76** as shown by an arrow B indicated by a one-dot chain line in FIGS. **6** and **7**.

(54) The bent flow path **96** is bent toward the Z(+) side in the Y-axis direction from the inflow portion **92** toward the jetting portion **94**. As an example, the bent flow path **96** is formed by a groove formed in the cap **34**, and a wall surface of the bent flow path **96** is formed as a liquid guiding surface. That is, the liquid flowing out from the outflow port **45** is guided in a direction toward the visual field region by the wall surface of the bent flow path **96**. In addition, according to the present embodiment in which the bent flow path **96** is formed of a groove, it is advantageous in a case where a thickness of a portion of the cap **34** where the bent flow path **96** is formed is not sufficient to form a through-hole.

(55) Next, the operation of the liquid jetting device **90** of the first embodiment will be described.

(56) By driving the liquid supply pump, the liquid supplied to the liquid supply tube **44** (see FIG. **3**) flows into the inflow portion **92** from the outflow port **45** in FIG. **6**. Then, the liquid is smoothly

guided to the jetting portion **94** by passing through the bent flow path **96**, and then is jetted from the jetting portion **94** toward the visual field region C as shown by the arrow B. Thus, the liquid can be supplied to an observation site located in the visual field region C. Since the cap **34** comprises the bent flow path **96** forming the liquid guiding surface, the liquid supply operation can be performed independently of the raising and lowering operation of the elevator **36** by the elevating operation lever **20** (see FIG. 1). Therefore, the raising and lowering operation of the elevator **36** and the liquid supply operation can be performed at the same time.

(57) As described above, according to the endoscope **10** of the embodiment, since the distal end portion main body **32** is provided with the outflow port **45** through which the liquid flows out, and the cap **34** is provided with the bent flow path **96** forming the liquid guiding surface, the treatment using the treatment tool can be performed while the liquid is supplied to the observation site. In other words, the liquid can be jetted toward the observation site without interfering with the operation of the treatment tool. Thus, for example, a bleeding part can be specified while washing it with liquid, and hemostasis can be performed using a treatment tool. In addition, endoscopic retrograde cholangiopancreatography can be performed using a treatment tool while washing away bile or fine gallstones with liquid.

(58) In the embodiment, although the liquid guiding surface is formed by the bent flow path **96**, the liquid guiding surface is not limited thereto, and may be formed of, for example, a linearly inclined flow path.

(59) In addition, in the embodiment, although the outflow port **45** is configured such that the outflow direction is directed in the direction along the longitudinal axis A direction, this configuration is not necessarily limited thereto, and the outflow port **45** may be configured such that the outflow direction is directed to the Z(+) side, the X(+) side, or the X(-) side, for example. However, according to the configuration of the outflow port **45** of the embodiment, in a case of the washing of the liquid supply tube **44** using washing liquid, which is performed by removing the cap **34** from the distal end portion main body **32**, the washing liquid can be made to flow out from the outflow port **45** toward the Y(+) side. That is, it is possible to prevent the washing liquid from splashing to the surroundings.

(60) Although the jetting portion **94** is open toward the visual field region C of the observation optical system **76**, it is more preferable that the jetting portion **94** is configured such that the jetting direction is directed to a position E separated from an outer surface of the observation optical system **76** in an optical axis D direction of the observation optical system **76**. By configuring the jetting portion **94** in this way, the liquid jetted from the jetting portion **94** can be reliably jetted to the position E separated from the observation optical system **76** in the optical axis D direction. In addition, the position E is more preferably a position corresponding to a focal point of the observation optical system **76**. Thus, a clear image of the observation site can be ensured.

(61) Further, according to the liquid jetting device **90** of the first embodiment, as shown in FIG. 8, the position of the jetting portion **94** in the longitudinal axis A direction is located on the base end side in the longitudinal axis A direction with respect to the position of the observation optical system **76** in the longitudinal axis A direction. Therefore, the liquid jetting device **90** of the first embodiment can obtain the following effects.

(62) That is, FIG. 9 shows an image F of the visual field region C (see FIGS. 6 and 7) displayed on the display **18** (see FIG. 1). Since the jetting portion **94** is disposed at the position shown in FIG. 8, the liquid jetted from the jetting portion **94** is jetted from an upper side of the image F toward an observation site H located at a center part of the image F, as shown by a thick arrow G in FIG. 9 on the image F.

(63) In FIG. 9, the elevator **36** and a treatment tool **100** are each shown by a two-dot chain line. The elevator **36** is not displayed on the image F, but is located on a right side of the image F. Then, the treatment tool **100** whose lead-out direction is changed by the raising and lowering operation of the elevator **36** is displayed on a right upper part of the image F indicated by reference numeral I in

a case where the elevator **36** is located in the elevating position, is displayed on a right center part of the image F indicated by reference numeral II in a case where the elevator **36** is located between the elevating position and the lying position, and is displayed on a right lower part of the image F indicated by reference numeral III in a case where the elevator **36** is located in the lying position. (64) Therefore, according to the liquid jetting device **90** of the first embodiment, the treatment tool **100** appearing on the right side of the image F can be guided to a target position of the observation site H by the elevator **36** while liquid is supplied from the upper side of the image F toward the observation site H of the image F.

(65) Although the jetting portion **94** shown in FIG. **8** is disposed substantially at the center of the cap **34** in the X-axis direction, the jetting portion **94** may be disposed in a position deviated in the X-axis direction from the position shown in FIG. **8** because the position of the jetting portion **94** need only be a position on the base end side in the longitudinal axis A direction with respect to the position of the observation optical system **76** in the longitudinal axis A direction. In this case, the jetting portion **94** is located in an optional position in a region J, which is formed on the upper side of the image F in FIG. **9** and is surrounded by a two-dot chain line, and the liquid can be jetted from the position to the observation site H.

(66) FIG. **10** is a top view of a distal end portion **112** having a liquid jetting device **110** of a second embodiment, and is a top view as viewed from the Z(+) side. FIG. **11** is a schematic cross-sectional view of the distal end portion **112** of FIG. **10** as viewed from the X(+) side. In the description of the liquid jetting device **110** of the second embodiment, the same or similar members as those of the liquid jetting device **90** of the first embodiment shown in FIGS. **6** to **8** will be denoted by the same reference numerals.

(67) As shown in FIGS. **10** to **11**, the liquid jetting device **110** of the second embodiment has: the liquid supply tube **44** having an outflow port **45A** that is open at a distal end thereof; and an inflow portion **114**, a jetting portion **116**, and a bent flow path **118** provided in the cap **34**.

(68) The outflow port **45A** is disposed on the Z(-) side of the distal end portion main body **32** with respect to the outflow port **45** shown in FIG. **6**. In addition, as shown in FIG. **11**, the outflow port **45A** is configured such that the outflow direction is directed in the direction along the longitudinal axis A direction.

(69) The inflow portion **114** provided in the cap **34** is provided at a position facing the outflow port **45A** in a state where the cap **34** is attached to the distal end portion main body **32**.

(70) As shown in FIG. **10**, the jetting portion **116** is provided on a distal end upper surface **34D** provided on the Y(+) side of the upper surface **34C** of the cap **34**. That is, a position of the jetting portion **116** in the longitudinal axis A direction is located on the distal end side in the longitudinal axis A direction with respect to a position of the observation optical system **76** in the longitudinal axis A direction. In addition, the jetting portion **116** is open toward the visual field region C of the observation optical system **76** as shown by an arrow K indicated by a one-dot chain line in FIG. **11**.

(71) The bent flow path **118** extends to the Y(+) side in the Y-axis direction from the inflow portion **114** toward the jetting portion **116**, and then is bent toward the Z(+) side.

(72) Next, the operation of the liquid jetting device **110** of the second embodiment will be described.

(73) The liquid supplied to the liquid supply tube **44** of FIG. **11** flows into the inflow portion **114** from the outflow port **45A**. Then, the liquid is smoothly guided to the jetting portion **116** by passing through the bent flow path **118**, and then is jetted from the jetting portion **116** toward the visual field region C as shown by the arrow K. Thus, the liquid can be supplied to an observation site located in the visual field region C. According to the liquid jetting device **110**, since the cap **34** comprises the inflow portion **114**, the jetting portion **116**, and the bent flow path **118**, the liquid supply operation can be performed independently of the raising and lowering operation of the elevator **36** by the elevating operation lever **20** (see FIG. **1**).

(74) Therefore, in the liquid jetting device **110** of the second embodiment as well as the liquid

jetting device **90** of the first embodiment, the treatment using the treatment tool can be performed while the liquid is supplied to the observation site.

(75) Further, according to the liquid jetting device **110** of the second embodiment, as shown in FIG. **10**, the position of the jetting portion **116** in the longitudinal axis A direction is located on the distal end side in the longitudinal axis A direction with respect to the position of the observation optical system **76** in the longitudinal axis A direction. Therefore, the liquid jetting device **110** of the second embodiment can obtain the following effects.

(76) That is, the liquid jetted from the jetting portion **116** is jetted from a lower side of the image F toward the observation site H located at the center part of the image F, as shown by a thick arrow L on the image F in FIG. **9**.

(77) Therefore, according to the liquid jetting device **110** of the second embodiment, the treatment tool **100** appearing on the right side of the image F can be guided to a target position of the observation site H by the elevator **36** while the liquid is supplied from the lower side of the image F toward the observation site H of the image F.

(78) Although the jetting portion **116** shown in FIG. **10** is disposed substantially at the center of the cap **34** in the X-axis direction, the jetting portion **116** may be disposed in a position deviated in the X-axis direction from the position shown in FIG. **10** because the position of the jetting portion **116** need only be a position on the distal end side in the longitudinal axis A direction with respect to the position of the observation optical system **76** in the longitudinal axis A direction. In this case, the jetting portion **116** is located in an optional position in a region M, which is formed on the lower side of the image F in FIG. **9** and is surrounded by a two-dot chain line, and the liquid can be jetted from the position to the observation site H.

(79) FIG. **12** is a top view of a distal end portion **122** having a liquid jetting device **120** of a third embodiment, and is a top view as viewed from the Z(+) side. In the description of the liquid jetting device **120** of the third embodiment, the same or similar members as those of the liquid jetting device **90** of the first embodiment shown in FIGS. **6** to **8** will be denoted by the same reference numerals.

(80) As shown in FIG. **12**, the liquid jetting device **120** has: the liquid supply tube **44** having an outflow port **45B** that is open at a distal end thereof; and an inflow portion **124**, a jetting portion **126**, and a bent flow path **128** provided in the cap **34**.

(81) The outflow port **45B** is disposed on a side surface of the distal end portion main body **32** on the X(+) side, and is configured such that the outflow direction is directed toward a direction on the X(+) side.

(82) The inflow portion **124** provided in the cap **34** is provided at a position facing the outflow port **45B** in a state where the cap **34** is attached to the distal end portion main body **32**.

(83) The jetting portion **126** is provided on a right upper surface **34E** provided on the X(+) side of the upper surface **34C** on the Z(+) side of the cap **34**. That is, the jetting portion **126** is disposed on a side opposite to the observation optical system **76** with the elevator housing portion **66** interposed therebetween. The jetting portion **126** is also open toward the visual field region C (see FIGS. **6** and **7**) of the observation optical system **76**.

(84) Although the detailed configuration of the bent flow path **128** is omitted, the bent flow path **128** extends to the Y(+) side in the Y-axis direction from the inflow portion **124** toward the jetting portion **126**, and then is bent toward the Z(+) side.

(85) Next, the operation of the liquid jetting device **120** will be described.

(86) The liquid supplied to the liquid supply tube **44** flows into the inflow portion **124** from the outflow port **45B** in FIG. **12**. Then, the liquid is smoothly guided to the jetting portion **126** by passing through the bent flow path **128**, and then is jetted from the jetting portion **126** toward the visual field region C. Thus, the liquid can be supplied to an observation site located in the visual field region C. According to the liquid jetting device **120**, since the cap **34** comprises the inflow portion **124**, the jetting portion **126**, and the bent flow path **128**, the liquid supply operation can be

performed independently of the raising and lowering operation of the elevator **36** by the elevating operation lever **20** (see FIG. **1**).

(87) Therefore, in the liquid jetting device **120** of the third embodiment as well as the liquid jetting device **90** of the first embodiment, the treatment using the treatment tool can be performed while the liquid is supplied to the observation site.

(88) Further, according to the liquid jetting device **120** of the third embodiment, the jetting portion **126** is disposed on the side opposite to the observation optical system **76** with the elevator housing portion **66** interposed therebetween. By disposing the jetting portion **126** in such a position, the liquid jetting device **120** of the third embodiment can obtain the following effects.

(89) That is, the liquid jetted from the jetting portion **126** is jetted from a right side of the image F toward the observation site H located at the center part of the image F, as shown by a thick arrow N on the image F in FIG. **9**.

(90) Therefore, according to the liquid jetting device **120** of the third embodiment, the treatment tool **100** appearing on the right side of the image F can be guided to a target position of the observation site H by the elevator **36** while the liquid is supplied from the right side of the image F toward the observation site H of the image F.

(91) Although the jetting portion **126** shown in FIG. **12** is disposed in the same position as the observation optical system **76** in the Y-axis direction, the jetting portion **126** may be disposed in a position deviated in the Y-axis direction (longitudinal axis A direction) from the position shown in FIG. **12** because the jetting portion **126** need only be disposed on the side opposite to the observation optical system **76** with the elevator housing portion **66** interposed therebetween. In this case, the jetting portion **126** is located in an optional position in a region P, which is formed on the right side of the image F in FIG. **9** and is surrounded by a two-dot chain line, and the liquid can be jetted from the position to the observation site H.

(92) FIG. **13** is a top view of a distal end portion **132** having a liquid jetting device **130** of a fourth embodiment, and is a top view as viewed from the Z(+) side. In addition, FIG. **14** is a side view of a left inner wall surface **34F** of the cap **34** as viewed from the X(+) side. In the description of the liquid jetting device **130** of the fourth embodiment, the same or similar members as those of the liquid jetting device **90** of the first embodiment shown in FIGS. **6** to **8** will be denoted by the same reference numerals.

(93) As shown in FIG. **13**, the liquid jetting device **130** has: the liquid supply tube **44** having an outflow port **45C** that is open at a distal end thereof; and an inflow portion **134**, a jetting portion **136**, and a bent flow path **138** provided in the cap **34**.

(94) The outflow port **45C** is disposed on a side surface of the distal end portion main body **32** on the X(-) side, and is configured such that the outflow direction is directed toward a direction on the X(-) side.

(95) The inflow portion **134** provided in the cap **34** is provided at a position facing the outflow port **45C** in a state where the cap **34** is attached to the distal end portion main body **32**.

(96) The jetting portion **136** is provided on a left upper surface **34G** provided on the X(-) side of the upper surface **34C** on the Z(-) side of the cap **34**. That is, the jetting portion **136** is disposed on a side opposite to the elevator housing portion **66** with the observation optical system **76** interposed therebetween. The jetting portion **136** is also open toward the visual field region C (see FIGS. **6** and **7**) of the observation optical system **76**.

(97) As shown in FIG. **14**, the bent flow path **138** is formed as a groove formed on the left inner wall surface **34F**. The bent flow path **138** extends to the Y(+) side in the Y-axis direction from the inflow portion **134** toward the jetting portion **136**, and then is bent toward the Z(+) side. The bent flow path **138** may be formed as a through-hole.

(98) Next, the operation of the liquid jetting device **130** will be described.

(99) The liquid supplied to the liquid supply tube **44** of FIG. **13** flows into the inflow portion **134** from the outflow port **45C**. Then, the liquid is smoothly guided to the jetting portion **136** by

passing through the bent flow path **138**, and then is jetted from the jetting portion **136** toward the visual field region C. Thus, the liquid can be supplied to an observation site located in the visual field region C. According to the liquid jetting device **130**, since the cap **34** comprises the inflow portion **134**, the jetting portion **136**, and the bent flow path **138**, the liquid supply operation can be performed independently of the raising and lowering operation of the elevator **36** by the elevating operation lever **20** (see FIG. 1).

(100) Therefore, in the liquid jetting device **130** of the fourth embodiment as well as the liquid jetting device **90** of the first embodiment, the treatment using the treatment tool can be performed while the liquid is supplied to the observation site.

(101) Further, according to the liquid jetting device **130** of the fourth embodiment, as shown in FIG. 13, the jetting portion **136** is disposed on the side opposite to the elevator housing portion **66** with the observation optical system **76** interposed therebetween. By disposing the jetting portion **136** in such a position, the liquid jetting device **130** of the fourth embodiment can obtain the following effects.

(102) That is, the liquid jetted from the jetting portion **136** is jetted from a left side of the image F toward the observation site H located at the center part of the image F, as shown by a thick arrow Q on the image F in FIG. 9.

(103) Therefore, according to the liquid jetting device **130** of the fourth embodiment, the treatment tool **100** appearing on the right side of the image F can be guided to a target position of the observation site H by the elevator **36** while the liquid is supplied from the left side of the image F toward the observation site H of the image F.

(104) Although the jetting portion **136** shown in FIG. 13 is disposed in the same position as the observation optical system **76** in the Y-axis direction, the jetting portion **136** may be disposed in a position deviated in the Y-axis direction (longitudinal axis A direction) from the position shown in FIG. 13 because the jetting portion **136** need only be disposed on the side opposite to the elevator housing portion **66** with the observation optical system **76** interposed therebetween. In this case, the jetting portion **136** is located in an optional position in a region R, which is formed on the left side of the image F in FIG. 9 and is surrounded by a two-dot chain line, and the liquid can be jetted from the position to the observation site H.

(105) In addition, according to the liquid jetting device **130** of the fourth embodiment, the liquid jetting direction (see arrow Q) can be opposed to the lead-out direction of the treatment tool **100** shown by an arrow S in FIG. 9. Therefore, there is an advantage in that the treatment tool **100** can be easily seen in comparison with the liquid jetting device **120** of the third embodiment in which the lead-out direction of the treatment tool **100** and the liquid jetting direction are the same on the image F.

(106) <Other Inventions>

(107) In the present invention, the liquid flowing out from the outflow port of the distal end portion main body is jetted into the visual field region by the liquid guiding surface provided in the cap. However, as another invention, as in the distal end portion main body **32** shown in FIG. 15, the distal end portion main body **32** may comprise a jetting portion **140** communicating with the liquid supply tube **44**. The distal end portion main body **32** comprises a fluid guiding surface that guides the liquid in the direction toward the visual field region between the jetting portion **140** and the liquid supply tube **44**. In addition, the jetting portion **140** is open toward the visual field region. Therefore, the liquid can be jetted from the jetting portion **140** of the distal end portion main body **32** into the visual field region. A position of the jetting portion **140** need only be disposed in a position exposed to the outside in a case where the cap **34** is mounted on the distal end portion main body **32**, and may be provided on the upper surface **68A** of the wall portion **68**, for example. In addition, in a case of the configuration of the distal end portion main body which does not use the cap, the jetting portion need only be disposed in an optional position.

(108) Although the present invention has been hitherto described, the present invention is not

limited to the above-mentioned examples, and various improvements and modifications may be made without departing from the scope of the present invention.

EXPLANATION OF REFERENCES

(109) **10**: endoscope **12**: endoscope system **14**: processor device **16**: light source device **18**: display **20**: elevating operation lever **22**: operation part **24**: insertion part **26**: flexible portion **28**: bending portion **30**: distal end portion **32**: distal end portion main body **34**: cap **34A**: opening window **34B**: end surface **34C**: upper surface **34D**: distal end upper surface **34E**: right upper surface **34F**: left inner wall surface **36**: elevator **36A**: base portion **38**: treatment tool channel **40**: elevating operation wire **42**: air and water supply tube **43**: joint pipe **44**: liquid supply tube **45**: outflow port **45A**: outflow port **45B**: outflow port **45C**: outflow port **46**: operation part main body **48**: grip portion **50**: bending-proof pipe **52**: universal cable **54**: light source connector **56**: electric connector **57**: air and water supply button **58**: air and water supply nozzle **59**: suction button **60**: treatment tool lead-out port **62**: angle knob **64**: treatment tool lead-in port **66**: elevator housing portion **68**: wall portion **68A**: upper surface **70**: wall portion **72**: housing chamber **74**: illumination optical system **76**: observation optical system **78**: elevating lever housing chamber **78A**: side surface **80**: elevating lever **80A**: lever portion **82**: through-hole **84**: through-hole **86**: rotary shaft **88**: hole **90**: liquid jetting device of first embodiment **92**: inflow portion **94**: jetting portion **96**: bent flow path **100**: treatment tool **110**: liquid jetting device of second embodiment **112**: distal end portion **114**: inflow portion **116**: jetting portion **118**: bent flow path **120**: liquid jetting device of third embodiment **122**: distal end portion **124**: inflow portion **126**: jetting portion **128**: bent flow path **130**: liquid jetting device of fourth embodiment **132**: distal end portion **134**: inflow portion **136**: jetting portion **138**: bent flow path **140**: jetting portion A: longitudinal axis C: visual field region D: optical axis E: position F: image H: observation site J: region M: region P: region R: region

Claims

1. An endoscope comprising: a distal end portion main body that is provided on a distal end side of an insertion part; an elevator housing portion that is provided in the distal end portion main body and is open in a first direction orthogonal to a longitudinal axis direction of the insertion part; an elevator that is rotatably provided in the elevator housing portion; an observation optical system that is disposed on a wall portion of the distal end portion main body adjacent to the elevator housing portion and that observes a visual field region in the first direction in which the elevator housing portion is open; an outflow port that is provided in the distal end portion main body and through which liquid flows out; and a cap that is attachably and detachably attached to the distal end portion main body and that has a liquid guiding surface that guides the liquid flowing out from the outflow port to a jetting portion of the cap in a direction toward the visual field region in a position separated from the observation optical system, wherein, in the longitudinal axis direction of the insertion part, the liquid guiding surface is disposed between the outflow port and a distal end of the cap, wherein the liquid guiding surface is a bent flow path formed by an inner wall surface of a groove formed in the cap, wherein the jetting portion is disposed on a base end of the cap in the longitudinal axis direction of the insertion part, wherein the jetting portion is configured such that a jetting direction of the liquid is away from the observation optical system.
2. The endoscope according to claim 1, wherein the cap has an inflow portion facing the outflow port in a state where the cap is attached to the distal end portion main body, the jetting portion that is open toward the visual field region, and a flow path connecting the inflow portion and the jetting portion to each other, and a wall surface of the flow path is formed as the liquid guiding surface.
3. The endoscope according to claim 2, wherein the jetting portion is configured such that a jetting direction is directed to a position separated from the observation optical system in an optical axis direction of the observation optical system.
4. The endoscope according to claim 2, wherein a position of the jetting portion in the longitudinal

axis direction is located on a base end side in the longitudinal axis direction with respect to a position of the observation optical system in the longitudinal axis direction.

5. The endoscope according to claim 2, wherein a position of the jetting portion in the longitudinal axis direction is located on a distal end side in the longitudinal axis direction with respect to a position of the observation optical system in the longitudinal axis direction.

6. The endoscope according to claim 2, wherein the elevator housing portion and the observation optical system are disposed adjacent to each other in a second direction orthogonal to the longitudinal axis direction and orthogonal to the first direction, and the jetting portion is disposed on a side opposite to the observation optical system with the elevator housing portion interposed therebetween.

7. The endoscope according to claim 2, wherein the elevator housing portion and the observation optical system are disposed adjacent to each other in a second direction orthogonal to the longitudinal axis direction and orthogonal to the first direction, and the jetting portion is disposed on a side opposite to the elevator housing portion with the observation optical system interposed therebetween.

8. The endoscope according to claim 4, wherein the outflow port is configured such that an outflow direction is directed in a direction along the longitudinal axis direction.

9. The endoscope according to claim 6, wherein the outflow port is configured such that an outflow direction is directed in a direction along the second direction.

10. The endoscope according to claim 1, wherein the liquid guiding surface is formed by an inner wall surface of a through-hole formed in the cap.
